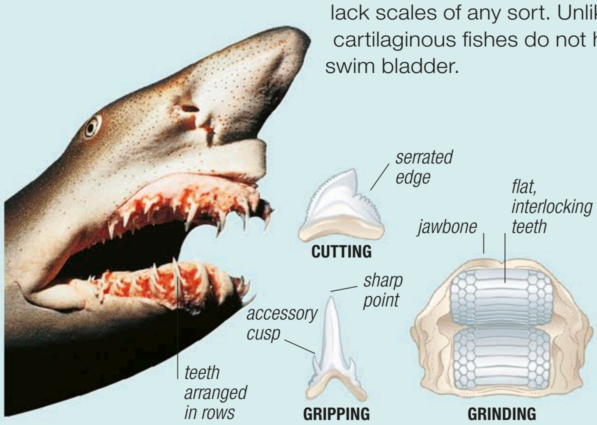
Cartilaginous fishes

PHYLUM	Chordata
CLASS	Chondrichthyes
ORDERS	14
FAMILIES	54
SPECIES	ABOUT 1,200

Although often described as primitive, cartilaginous fishes include some of the largest and most successful of all marine predators. They are divided into 3 groups: sharks; skates and rays; and a group of deep-sea fishes called chimeras. All possess a skeleton made of cartilage (rather than bone), specialized teeth that are replaced throughout their lifetime, and skin covered with small, dense, toothlike scales. Most cartilaginous fishes are marine, but some sharks, skates, and rays enter freshwater, and certain tropical species live exclusively in freshwater.

Anatomy

All the fishes in this group have an internal skeleton made of tough, flexible cartilage. This may be strengthened by mineral deposits, and some species have hard, bonelike dorsal spines. However, the skeleton of a cartilaginous fish is much more flexible than that of a bony fish (see p.486), which has a much higher mineral content. On land, a cartilage skeleton would not be rigid enough to support the weight of a large animal, but in water—which has a much higher density than air—cartilage provides an effective skeleton for animals that are up to 33 ft (10 m) long. Most cartilaginous fishes have skin covered by thousands of interlocking scales, called placoid scales or dermal denticles (see p.468). These have a similar composition to teeth and give the skin a sandpaper-like texture. Some rays have additional large, thornlike scales, while chimeras lack scales of any sort. Unlike bony fishes, cartilaginous fishes do not have a gas-filled swim bladder.



TEETH
Cartilaginous fishes have teeth that are shaped to suit their diet. Serrated teeth are used for cutting, while pointed teeth are used to hold prey. Many rays have flat teeth for grinding food. As teeth are lost, they are replaced by new ones growing behind them.

GILL OPENINGS
Most sharks and rays have 5 pairs of gill openings (a few have 6–7) while chimeras have only one. When water enters the mouth, the gill slits are closed. As the water passes out of the open gill slits, the mouth is held closed.

Senses

Cartilaginous fishes have acute senses that can be used to locate prey, even if it is a long way off or buried by sediment. All species have a system of pores, called the ampullae of Lorenzini, which they use to detect the weak electrical signals emitted by other animals. Most also have an effective lateral-line system (see p.469) that responds to very small vibrations, as well as acute vision and a sense of smell that can detect odors in even the weakest solutions.

ELECTRICAL SIGNALS
The ampullae of Lorenzini (seen here on the snout of an oceanic whitetip shark) are deep pores connected to electro-receptive nerve endings.

Feeding

All cartilaginous fishes are carnivorous; however, plankton feeders also take in microscopic plants. The greater part of the diet of most species consists of live prey, including other fishes, invertebrates, and occasionally marine mammals. Most species also feed on the remains of dead animals, although only a few species rely solely on carrion. Several of the largest sharks and rays are filter feeders. The teeth of these species have been reduced in size, and stiff projections in the gill arches have developed into sievelike organs for straining small animals out of water.

FILTER FEEDING
Whale sharks drift slowly through water, catching plankton, small fishes, and squid. They live in warm tropical waters.

SWIMMING AND BUOYANCY
Cartilaginous fishes, such as the hammer-head shark, have an oil-rich liver that adds to their buoyancy. However, most are still negatively buoyant and must keep swimming to avoid sinking.

Reproduction

In all cartilaginous fishes, fertilization occurs internally. The male passes sperm into the female's cloaca via a modified pelvic fin. The young are produced by one of 3 processes. The females of some species release leathery egg cases. In other species, the young hatch from the egg inside the female's body and are then released alive. In the remaining species, the young develop inside a placenta-like structure, from which they have a direct nutritional connection with the female. In all cases, there is no larval stage; instead, the young are born as miniature versions of the adults.

LIVE BIRTH
Young lemon sharks are nourished via a yolk sac placenta and are born tail-first .

EGG CASE
This egg case contains the embryo of a small-spotted catshark. It is anchored to seaweed by tendrils on the outside.